EXECUTIVE SUMMARY
SOUTHERN HEMISPHERE SPACE STUDIES PROGRAM 2023



JOURNEY TO THE MOON

TECHNOLOGICAL AND LEGAL PERSPECTIVES FOR SUSTAINABLE HUMAN PRESENCE



INTERNATIONAL SPACE UNIVERSITY

University of South Australia

VISION

A sustainable lunar ecosystem with permanent human presence on the Moon, poised to be the starting point for humanity's expansion into the Solar System.

OBJECTIVES

- Describe the current state of national, international, and commercial space efforts and relevant history
- Recommend a roadmap toward sustainable human presence, focused on key lunar resources and technologies
- Recommend a comprehensive international legal framework that enhances and incentivizes cooperation between parties active on the Moon

OVERVIEW

Today's challenges of long-term sustainable life in the harsh environment of outer space is driven by the combined efforts of private and public space organizations. This report explores different recommendations that address the complex intricacies of future lunar activities, presents the technological issues that have to be tackled in the coming years, and illustrates the major concerns that derive from these issues.

The focus has been on the handling of the key aspects that relate most strongly to the physical sustainability of a lunar settlement: power, in-situ resource utilization (ISRU) and human habitation. In terms of timeline, this team project addresses the challenge of sustainability by considering the following roadmap: near-term (~ 2030), mid-term (~ 2040) and long-term (~2050).

This report acknowledges the importance of international co-operation in sustaining a long-term human exploration effort in outer space and beyond. The current system of international law is insufficiently primed to tackle specific problems in the emerging lunar domain. Accordingly, this report further recommends a novel two-tiered international legal framework and, in furtherance, prescribes a suggested model.

FUTURE

2028 - CHINA
Chang'e 8
Testing ISRU

2028 - RUSSIA
Luna 28
Sample return from South Pole

2026 - CHINA
Chang'e 7
Survey and exploration of South Pole

2026 - CHINA
Chang'e 6
Sample return from South Pole

2025 - USA
Artemis III +
First Crew landing since Apollo

2025 - RUSSIA
Luna 27
Lander to investigate site for lunar base

2024 - USA
Artemis II
Crewed flight test

2023 - RUSSIA
Luna 25
Lander to investigate South Pole

2022 - USA
Artemis I
Uncrewed Moon orbiter

2020 - CHINA
Chang'e 5
Lunar sample return

2019 - INDIA
Chandrayan-2
Orbiter Mission (in progress)

2019 - CHINA
Chang'e 4
Yutu-2 rover on far side of Moon

2013 - CHINA
Chang'e 3
Yutu rover on Lunar surface, first in 37 years

2008 - INDIA
Chandrayaan 1
Discovery of water on lunar surface

2007 - CHINA
Chang'e 1
First Chinese Robotic mission

2003 - EUR
SMART 1
3D X-ray and IR imagery of lunar surface

1990 - JAPAN
Hiten
Japan becomes third nation to place an object on Moon

1972 - USA
Apollo 17
Final mission of Apollo

1970 - USSR
Lunokhod 1
First rover landing on Moon

1969 - USA
Apollo 11
First human landing

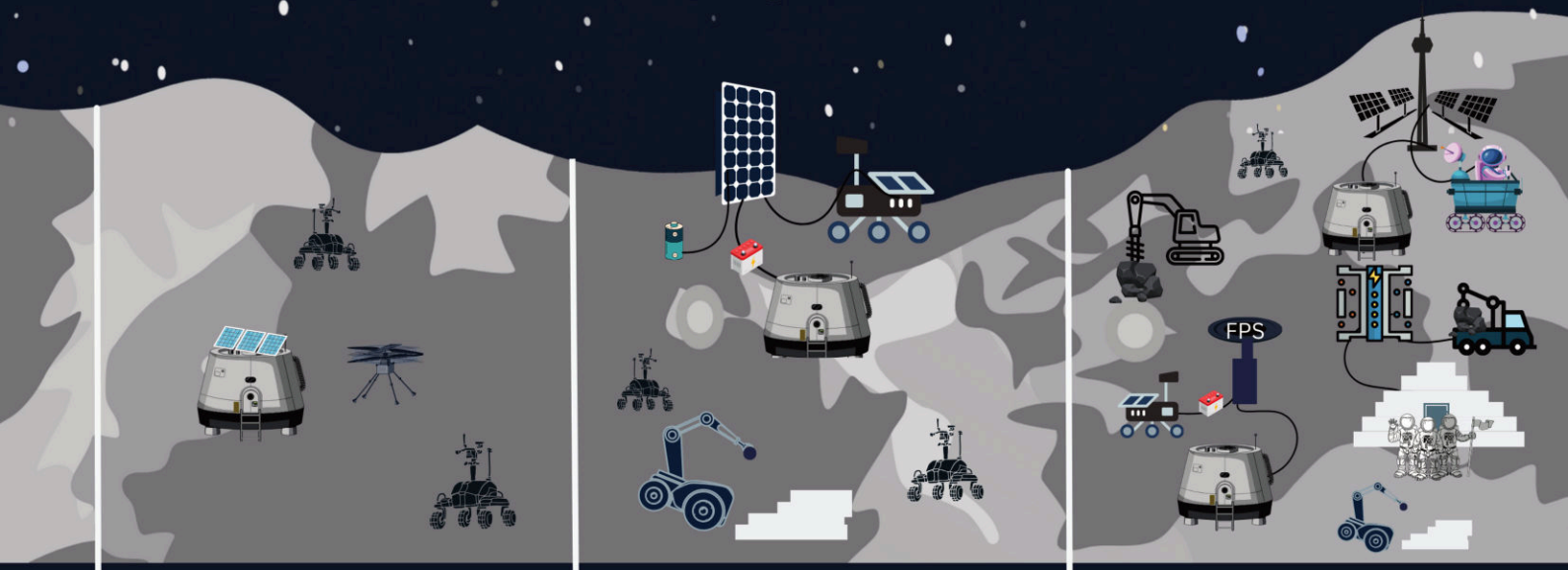
1966 - USSR
Luna 9-10
Luna 9 – First soft landing on Moon
Luna 10 – First orbit around the Moon

1959 - USSR
Luna 2-3
Luna 2 – First probe to impact Moon surface
Luna 3 – First photo of far side of the Moon

This timeline presents significant lunar missions as well as recent and proposed future missions. The recommendations from this project will begin influencing lunar activities 2023 onward.

A ROADMAP FOR HUMAN

Near Term → Transition → Mid Term



2023

- Radioactive thermoelectric generators (RTGs)
- Radioactive heating units (RHUs)
- Integrated solar cells & batteries
- Integrated power management
- Robotic exploration
- In space manufacturing (ISM) trial
- Gateway
- Single Artemis Base Camp (ABC)
- Quantification of resources

2035

- Solar station
- Li-ion batteries as backup
- Power distribution established
- Increased robotic exploration
- In space manufacturing validated
- Resupply from Gateway
- Open loop system
- Application of ISRU for lunar regolith shielding

2040

- Concentrated solar power
- Fission power system (FPS)
- Hydrogen fuel cells
- Independent micro grids
- Advanced robotic for in-situ exploitation and transportation
- Expansion of ABC
- Hybrid life support system with water and oxygen from ISRU and Earth

3

○ Power □ In-situ resource utilization (ISRU) & manufacturing △ Habitation

HABITATION ON THE MOON

Transition

Long Term



2045

Expanded solar production

Power management system

Scientific research modules

2050

Smart automated power grid

ISRU of water for scalable hydrogen fuel cells

Material extraction and production by ISRU

ISRU for solar panel production

Reduced dependence on Earth

Scalable solar farm from ISRU

Hydrogen fuel cells with ISRU manufactured Na-ion batteries

Automated transportation network of people and goods

Full sustainable manufacturing

Large south pole habitation with investigation of alternative habitation

Blue: use resources brought from Earth

White: use recourses from the Moon

LEGAL FRAMEWORK

We believe the existing framework is insufficiently primed to tackle the most pressing contemporary and future issues surrounding the expected activities on the Moon. Thus, we are presenting a two-tiered legal framework, comprising of:

- A 'living document'- a treaty which is signed and ratified by states; and
- A mechanism that gives stakeholder-voiced advisory to be formulated as Agreements under the Treaty

LUNAR ACTIVITIES TREATY (LAT)

- Membership limited to nation states
- Modelled on the Antarctic Treaty System and ISS Intergovernmental Agreement
- Inherently a 'living document'; the Treaty is iteratively updated through addition of Agreements that resolve contemporary topics

COMMITTEE OVERSEEING LUNAR ACTIVITIES (COLA)

Secretariat	Decision Making Council (DMC)	Dispute Resolution Body (DRB)
• Administrative arm • Handles the day to day activity • Expert committees which advise the Decision Making Council • Refers matters of contention to Dispute Resolution Body	Gives advisories to be ratified by the Treaty Body Five entities (stakeholder voices): • States • Corporates • Employees (Unions) • Non-government organization • Academia	Consultations Panel — Arbitration Appellate body

CONCLUSIONS

- Potential geopolitical concerns could arise as a result of the competition between Artemis and ILRS
- To achieve sustainable human presence on the Moon, we have proposed an approach based on technological development and the creation of a new legal framework, based on the Third World Approaches to International Law (TWAIL)
- With the advancement of solar power efficiency and production, additive manufacturing, habitation, and life support systems, the long-term future of sustainable presence and a thriving lunar economy can be achieved
- The suggested legal framework, comprising LAT and COLA, is not a static model that addresses all possible points of contention. As a living document, it can be reviewed annually and adapted to contemporary demands
- The code of conduct created by the COLA members helps ensure safe, sustainable lunar activity today, and for preserving it into the future



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